

Economy Feasibility of Methanol Production

CHE 396: Mini-Project 3

Group #1

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Introduction

Overview

This report presents a comprehensive economic feasibility analysis for a new methanol production plant via carbon hydrogenation. The objective of this study is to determine the financial viability of constructing and operating a facility that converts carbon feedstock and hydrogen into methanol, which is a key chemical process used in fuels, solvents, and polymer precursors.

This study integrates process design data, including a Process Flow Diagram (PFD), a material balance table, equipment specifications, and a utilities list. Using this information, both technical and financial analyses are performed to estimate total capital investment, production costs, revenues, and key profitability indicators, including ROI, NPV, and payback period by hand calculation and through Aspen PLUS. Sensitivity analyses are conducted to further evaluate the project's robustness to changes in key economic variables, including feedstock cost, utility pricing, and product value.

The report concludes with an AI-assisted evaluation that explores the potential of modern computational tools in performing cost estimation and financial forecasting, comparing their output with conventional engineering methods.

Design Charter

Design Plan

Establish a purpose and design basis for this operation. Conduct extensive research about the market. Create a quantitative analysis on the statistical information

Objective

Determine viability of constructing and operating a facility

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Design Plan

Establish a purpose and design basis for this operation. Conduct extensive research about the market. Present a quantitative analysis on the statistical information

Executive Summary

This report provides a comprehensive economic evaluation of a proposed methanol production plant that converts carbon monoxide and hydrogen into methanol. Using the provided process flow diagram, equipment specifications, utilities, and material balances, the financial feasibility was assessed.

Capital investment was determined using the order-of magnitude method and detailed equipment costing through hand calculations and Aspen PLUS. Total Capital Cost, which includes ISBL, OSBL, engineering costs, and contingency was calculated to establish the required investment. The operating costs were divided into variable costs based on raw material and utility prices, and fixed costs assumed 8,000 operating hours.

Economic performance was measured through annual revenue, gross profit, payback period, and ROI. The Net Profit Value was calculated at 14% discount rate using both MACRS-5 and MACRS-7, which was used to find the Discounted Cash Flow Rate of Return. The results show minimal differences between the two depreciation schemes, with both yielding positive profit rates and returns.

The sensitivity analysis was performed on the capital cost, utilities, raw material cost, and product price. The analysis demonstrated that the revenue and variable operating costs have the greatest impact on profitability, whereas the capital and fixed costs were less influential.

The study includes an AI-generated economic assessment, which demonstrated the tool's accuracy in calculations and real-world applicability.

Overall, this study indicates that the methanol production plant is economically viable, with strong long-term profitability and resilience under reasonable cost fluctuations.

Capital Cost Estimation

Order of Magnitude Estimate

An order-of-magnitude estimate of the installed capital cost for the methanol production plant was developed using the capacity-factor (six-tenths) rule. The reference basis was the Methanex Geismar 3 methanol plant in Louisiana (Methanex, 2019), which has a capacity of 1.8 million MT per year and a reported capital cost of approximately \$1.35 billion USD. The cost was scaled using the following relationship:

$$C = C_{ref} * \left(\frac{S}{S_{ref}} \right)^n$$

where C_{ref} is the reference cost (\$1.3 billion), S_{ref} is the reference capacity ($1.8 * 10^6$ tonnes/year), S is the proposed capacity ($8.8 * 10^5$ tons/year), and $n = 0.6$. The estimated capital cost was calculated to be 0.88 billion USD.

Detailed Equipment Cost Evaluation

Using Aspen Plus Economic Analyzer to determine a “Total Project Capital Cost” (TPCC) allows for a fair comparison between hand calculations and software-based calculations. A few key differences between Aspen and hand calculations are that Aspen Plus uses a pricing basis from quarter 1 of 2022, which can result in some marginal pricing discrepancies for both equipment and utilities. Additionally, Aspen Plus requires very little information to calculate the

Economic Feasibility

cost of each unit, such as heat-exchange area, number of stages, height, and diameter, among others. Aspen facilitates ease of use for a rough breakdown and estimation of the project at hand. In the case of this project, *Methanol Synthesis via Carbon Hydrogenation*, certain units, such as the reactor, can be isolated from the TPCC for reasons including the complex reactor design and the lack of specifications provided for the project.

The units that contributed to the TPCC were E-300, E-300*, E-301, E-302, T-300, K-300, D-300, and D-301. As for the heating units, the E-300, E-300*, and E-301 were modeled as TEMA shell-and-tube heat exchangers with their respective heat-exchanging areas of 67 m², 3,020 m², and 3300 m². Additionally, E-302 was modeled as a kettle reboiler, as it was the bottom product of the distillation column (T-300) with a heat-exchange area of 6,130 m². Moreover, the T-300 was modeled as a trayed tower with a column diameter of 7.9 m, a column height of 12.7 m, 23 stages, and a design pressure of 55 bar, which is slightly higher than the stream pressure entering the unit. The compressor K-300 was modeled as a horizontal centrifugal compressor with a gas flow rate of 16230 LPM. Finally, the separators, D-300 and D-301, were both modeled as vertical process vessels with both units having a diameter of 3.2 m and a height of 7 m. D-300 and D-301 followed the same design pressure of 50 bar, which is similar to that of T-300, being higher than the inlet pressure of the stream into D-300. The higher-pressure input is due to it being a design pressure, which is typically higher than the actual pressure entering the unit, as conditions are different inside.

As for the cost breakdown of each unit and the utilities recorded, Aspen gave the following:

Table 1: Costs of Equipment and Utilities Calculated by Aspen

Equipment ID	Equipment Cost	Total Direct Cost
E-300	\$37,400	\$153,600
E-300*	\$831,800	\$1,303,800
E-301	\$1,303,800	\$1,303,800
E-302	\$1,656,800	\$2,198,800
T-300	\$5,766,500	\$9,103,800
K-300	\$977,700	\$1,193,400
D-300	\$195,200	\$454,800
D-301	\$195,200	\$454,800
Utilities Cost		\$114,316.84
Total Operating Cost		\$2,566,422.19
Total Capital Cost		\$31,002,537.41

The use of Aspen Plus extends beyond process simulation, and the Economic Analyzer is one of the many tools available within Aspen Plus. The ease of using Aspen for a cost analysis consists of its simulation integration, time efficiency, consistency, and standardization, as well as scenario and sensitivity analysis, and ultimately, detailed output.

Aspen's simulation integration enables data to be pulled directly from streams, equipment, and specifications within the simulation. The benefit of this automation is the reduction of manual user inputs and the minimization of possible errors associated with manually inputting data. Moreover, automation enables quick changes to the process and reevaluation of the cost estimation in mere seconds. This eliminates the need for strenuous, repeated hand

calculations, ensuring that the estimation performed through Aspen is based on the simulation or inputs from the process design, conditions, and specifications.

When it comes to time efficiency, the use of Aspen's Economic Analyzer enables quick estimates and a plug-and-play solution for various projects and scenarios. This can be especially useful in a large plant setting with multiple utilities and equipment. The software is also very flexible, allowing numerous inputs and design decisions that impact the final cost estimation. Flexible in the sense that inputs can be changed with ease, allowing quick answers to design questions and constraints.

The consistency and Standardization of the Aspen Plus Economic Analyzer enable the estimation to be reflected on a single basis. The Economic Analyzer uses a pricing basis from quarter 1 of 2022, allowing for the numbers to be reflected and adjusted to the values of 2022. This allows the user to be confident in the consistency of the cost estimation and enables the use of a cost escalation to convert to the current year of design. Aspen utilizes a wide range of databases and sources for its estimations, including, but not limited to, the Chemical Engineering Plant Cost Index and existing vendor data.

The use of Aspen enables scenario and sensitivity analysis to be conducted without additional input. The ease of performing a "what-if" analysis, for example, effects of design changes, process conditions, equipment sizing, and the type of equipment and utilities. These factors all have a significant impact on the cost estimation, allowing the project designer to test the capabilities of their design and identify the maximums and minimums they can achieve. This allows the project to be compared to other existing projects or other formats of the current project, enabling a weighing of the differences in design and long-term investment.

Finally, the detailed output that Aspen provides allows the user to extract important numbers and provides a breakdown of how they were calculated. Aspen breaks down into ISBL, OSBL, fixed capital, working capital, and operating cost. This allows the user to avoid these strenuous calculations and tailor the entire estimation to their intended specifications, without the headache of repeating scenarios and calculations on the same project.

Contrarily, Aspen has cons to go with its many pros, such as data dependence and accuracy, limited for conceptual and novel designs, and a lack of “engineering judgement.” These constraints highlight the limitations that Aspen Plus encounters when performing a cost estimation.

The data dependence and accuracy of the estimation are the biggest constraints of the software. Aspen Plus has a heavy reliance on internal databases and default settings, which may not accurately reflect the project. Aspen’s internal basis may not align with existing industry prices or those from specific equipment vendors. The cost estimation may differ drastically if the cost index for the year and equipment type does not match the market trend and pricing. The limitation of the cost being from quarter 1 of 2022 can lead to some misleading information as the years pass, and factors such as supply, demand, taxation, etc., have a significant impact on market pricing.

The use of the quick cost estimation by Aspen isn’t ideal for non-conceptual or novel designs. For example, the Methanol Synthesis via carbon hydrogenation requires the use of a plug flow reactor; however, Aspen Plus Economic Analyzer didn’t have an option for reactor estimation due to the complex design and other factors. The innovative projects that regulate non-standard equipment may prove to be difficult for Aspen to estimate a price due to indirect workarounds being used.

Finally, the use of the Economic Analyzer will always lack the “engineering judgement.” Aspen typically assumes a standard condition; however, this does not account for other engineering factors, such as engineering design, labor rates, local taxes, and environmental permits for the plant. The lack of critical thinking allows Aspen to water down the analysis to a point where it becomes feasible and possible. Rather than a realistic and accurate measurement of a project, whether large or small.

Breakdown of Capital Investment

Purchased equipment costs were calculated for each major unit using the following correlation (Towler & Sinnott, 2013):

$$C_e = a + bS^n$$

where C_e is the purchased cost on a 210 U.S. Gulf Coast Basis, S is the size parameter of each equipment, and $a, b, \text{ and } n$ are equipment specific constants. Each purchased equipment cost was then multiplied by its corresponding installation factor to obtain the installed cost. The total installed cost was then escalated to the current year using the cost-index relation:

$$\text{Cost in current year} = \text{Cost in base year} * \frac{CEPCI_{\text{current}}}{CEPCI_{\text{base}}}$$

A detailed cost breakdown for all major process equipment is presented in Table 2:

Table 2: Purchase Cost of Equipment

Equipment	S	a	b	n	Cost	Installation Factor	Installed Cost
E-300 (Exchanger)	67.00	2800	54	1.20	\$11,188.48	3.5	\$39,159.70
E-300* (Steam Generator)	3020.00	2800	54	1.20	\$812,645.35	3.5	\$2,844,258.72
E-301 (Condenser)	3300.00	2800	54	1.20	\$903,562.89	3.5	\$3,162,470.13
E-302 (Reboiler)	6130.00	29000	400	0.90	\$1,054,119.32	3.5	\$3,689,417.62
T-300 (Distillation Column)	50431.36	17400	79	0.85	\$802,495.81	4.0	\$3,209,983.23
Sieve Trays	23.00	130	440	1.80	\$420,733.09	1.3	\$546,953.02
K-300 (Compressor)	973.80	4450	57	0.80	\$18,466.86	2.5	\$46,167.14
D-300 (Liquid-Vapor Separator)	3237.10	17400	79	0.85	\$93,478.06	4.0	\$373,912.26
D-301 (Liquid-Vapor Separator)	3237.10	17400	79	0.85	\$93,478.06	4.0	\$373,912.26

The resulting installed costs for all equipment were summed up to obtain the ISBL cost of \$23.84 million USD. The Inside Battery Limits (ISBL) includes all process units directly associated with the methanol synthesis system.

To account for costs beyond the process boundary, the Outside Battery Limits (OSBL) cost was estimated as 40% of the ISBL value. This percentage reflects the additional investment required for utility generation. Accordingly, OSBL was calculated at \$9.54 million USD.

Additional indirect costs were estimated as percentages of the ISBL. Engineering costs were taken as 10% of ISBL (\$2.38 million) to cover detailed design and construction. A 10%

contingency (\$2.38 million) was included to allow for design uncertainties and unpriced items. Working capital, which covers operating supplies and start-up expenses, was estimated as 15% of ISBL (\$3.58 million), representing a mid-range value commonly used for chemical process projects. A detailed cost breakdown is shown in Table 3:

<u>Table 3: Breakdown of Total Capital Cost</u>	
Component	Cost in USD (MM)
ISBL	\$23.84
OSBL	\$9.54
ECC	\$2.38
Contingency	\$2.38
WC	\$3.58
Total Capital Cost	\$41.72

Operating Cost Analysis

Variable Costs

The Variable Cost of Production (VCOP) was determined using the hourly material and utility flow rates provided. Each flow rate was multiplied by the corresponding unit price of material or utility, obtained from current industrial references. The total hourly cost was then converted to an annual basis, assuming 8,000 operating hours per year, consistent with standard chemical plant operation schedules.

As shown in Table 4, hydrogen represents the major cost contributor due to its high price and consumption rate, followed by LP steam and carbon dioxide. Cooling water, electricity, and waste treatment contribute relatively small portions to the overall variable cost.

Table 4: Raw Material and Utility Costs for VCOP Calculation

Material/Utility	Amount (MT/hr or kWh)	Price (\$/ton or \$/kWh)	Cost (\$/hr)	Total VCOP (\$/hr)	Total VCOP (\$/year)
H2	21	1,000.00	21,000.00	24,322.63	194,581,016.00
CO2	151.5	5.00	757.50		
LP Steam	150	11.45	1,717.50		
Cooling Water	7330	0.10	733.00		
Electricity	150	0.14	21.00		
Waste	0.3	1.59	0.48		
Waste Water	62.1	1.50	93.15		

The resulting VCOP is \$194.6 million per year. This estimate includes all raw materials, utilities, and waste handling costs. Hydrogen accounts for approximately 85% of the total variable cost, indicating that production cost is highly sensitive to hydrogen pricing.

Fixed Costs

The Fixed Cost of Production (FCOP) represents all annual expenses that remain constant, regardless of the quantity of products manufactured. These costs are tied to maintaining plant operations, personnel, and equipment rather than to the amount of methanol produced. The FCOP was determined after estimating the VCOP values for all raw materials and utilities. The VCOP data was first extracted from the hand calculations and then used to calculate the FCOP.

Standard process-economic correlations were used to calculate the operating labor, maintenance, depreciation, insurance, taxes, and administrative overhead (Towler & Sinnott, 2013). Each of these items was estimated as a percentage of the Fixed Capital Investment (FCI). The total FCI for the process was approximately \$38.14 million, which served as the basis for calculating each fixed-cost component. Taxes and insurance were estimated at 1.5% of the FCI, resulting in approximately \$0.57 million per year, while depreciation was set at 10% of the FCI, or roughly \$3.8 million per year. Interest in debt financing was assumed to be zero since no external loans were considered in this project. After summing all contributions, the total Fixed Cost of Production (FCOP) was found to be approximately \$6.65 million per year. This represents the annual fixed operating expense required to sustain plant operations, including overhead, maintenance, and administrative costs, regardless of fluctuating production levels.

Profitability Analysis

Revenue Estimation

The total annual revenue was calculated from the methanol production rate and the assumed selling price of methanol. The production rate obtained from the material balance was multiplied by the current market price of methanol (\$802/MT) (Corporation, 2025) and the annual operating hours (8000 hr/year). The resulting revenue was compared to the total manufacturing cost, which included both the Variable Cost of Production (VCOP) and the Fixed Cost of Production (FCOP). The gross profit was then determined by subtracting these combined

costs from the total revenue (Towler & Sinnott, 2013). This metric provides a direct measure of the plant's operational profitability before taxes or financing.

Payback and Return

The Simple Payback Period and Return on Investment (ROI) were calculated to evaluate the profitability of the methanol production process. ROI measures the annual pre-tax return as a percentage of the total capital invested, while the payback period represents the time required to recover the total investment from cumulative cash flows.

These values were derived using the total capital cost (TCC), working capital (WC), annual revenues, and annual operating costs. The TCC was \$41.72 million, and the WC was \$3.58 million, resulting in a total capital investment (TCI) of \$ 45.3 million. The total annual revenue was calculated from the selling price of methanol and the production rate, resulting in \$705.76 million per year. Meanwhile, the combined variable and fixed operating costs (VCOP + FCOP) totaled \$201.23 million per year. The annual pre-tax cash flow (PTCF) was determined by subtracting these operating costs from the total revenue, resulting in a net amount of \$504.53 million.

This PTCF was then used to compute the ROI and simple payback period. The ROI was calculated by dividing the annual PTCF by the TCI and then multiplying the result by 100 to express the value as a percentage. This results in a ROI of 1114.3%. This suggests a highly profitable design under the given assumptions, with a strong cash flow relative to the capital cost. The simple payback period was calculated by dividing the TCI by the annual PTCF,

yielding 2.11 years. This means that the plant would recover its total investment in just over two years of operation. Overall, the analysis suggests that the process and design are economically attractive, offering a strong return on investment and low capital risk.

Cash Flow and Investment Evaluation

Net Present Value

The NPV was calculated under two depreciation schemes, MACRS-5 and MACRS-7, to assess the influence of the depreciation method. The interest rate that was kept in mind was 14%, and the plant operated for a total of 22 years, which is typical for a plant of this production size and equipment types (Towler & Sinnott, 2013).

The specific key parameters that were kept in mind include the Inside Battery Limits Capital Cost (ISBL), Variable Capital Cost (VCOP), Fixed Operating Cost (FCOP), and Revenue. Revenue is based on the selling price of the product, specifically methanol, which is \$802 USD per metric ton. (Methanex, 2025).

The process of determining the base NPV value at an interest rate of 14% involved creating a table of cash flows across the 22 years of operation. The initial 2 years were set aside for creating the production plant, in which the entirety of the Total Capital Cost (TCC) will be used in the Pre-Tax Cash Flow (PTCF). The TCC value was calculated to be \$ 41.7 million,

where 25% of that amount was used in Year 1 and 75% in Year 2. Year 3 is considered the first year of operation, during which only 50% of the operating capacity is utilized.

Year 3:

$$\text{Pre - Tax Cash Flow} = \frac{\text{Revenue}}{2} - \frac{\text{VCOP}}{2} - \text{FCOP}$$

From Year 4 to Year 22, the value of the PTCF is the difference between the Revenue and the Capital Cost of Production (CCOP), the sum of VCOP and FCOP. From this, the Taxable Income is calculated by finding the difference between the Pre-Tax Cash flow and the Depreciation Value.

Taxable Income:

$$\text{Taxable Income} = \text{PTCF} - \text{MACRS} * (\text{TCC} - 0.0857 * \text{TCC})$$

The tax that is assumed is 35% of the taxable income (Towler & Sinnott, 2013), which is taken from the taxable income to calculate the Cash Flow (CF). From this, the Present Value (PV) will be calculated by considering the Discount Factor (DF).

$$DF = \frac{1}{(1 + i)^n}$$

i = Interest Rate
n = Year

Finally, the Net Present Value (NPV) was calculated by summing the Present Value of each year.

Tables 5 and 6 summarize the detailed cash flow calculations for each depreciation schedule.

Table 5: NPV Calculation using MACRS-5

Years	Pre-Tax Cash Flow (MM)	Depreciation 5	Taxable Income 5	Tax 5	Cash Flow 5	Discount Factor	Present Value of CF 5	NPV 5
1	-10.43	0.00	-10.43	0.00	-10.43	0.88	-9.15	-9.15
2	-31.29	0.00	-31.29	0.00	-31.29	0.77	-24.07	-33.22
3	248.94	7.63	241.31	0.00	248.94	0.67	168.03	134.81
4	504.53	12.21	492.33	84.46	420.07	0.59	248.72	383.52
5	504.53	7.32	497.21	172.31	332.22	0.52	172.54	556.07
6	504.53	4.39	500.14	174.02	330.51	0.46	150.58	706.64
7	504.53	4.39	500.14	175.05	329.48	0.40	131.67	838.31
8	504.53	2.20	502.34	175.05	329.48	0.35	115.50	953.82
9	504.53	0.00	504.53	175.82	328.71	0.31	101.08	1054.90
10	504.53	0.00	504.53	176.59	327.95	0.27	88.46	1143.36
11	504.53	0.00	504.53	176.59	327.95	0.24	77.60	1220.96
12	504.53	0.00	504.53	176.59	327.95	0.21	68.07	1289.03
13	504.53	0.00	504.53	176.59	327.95	0.18	59.71	1348.74
14	504.53	0.00	504.53	176.59	327.95	0.16	52.38	1401.11
15	504.53	0.00	504.53	176.59	327.95	0.14	45.94	1447.06
16	504.53	0.00	504.53	176.59	327.95	0.12	40.30	1487.36
17	504.53	0.00	504.53	176.59	327.95	0.11	35.35	1522.71
18	504.53	0.00	504.53	176.59	327.95	0.09	31.01	1553.72
19	504.53	0.00	504.53	176.59	327.95	0.08	27.20	1580.92
20	504.53	0.00	504.53	176.59	327.95	0.07	23.86	1604.79
21	504.53	0.00	504.53	176.59	327.95	0.06	20.93	1625.72
22	504.53	0.00	504.53	176.59	327.95	0.06	18.36	1644.08

Table 6: PV Calculation using MACRS-7

Years	Pre-Tax Cash Flow (MM)	Depreciation 7	Taxable Income 7	Tax 7	Cash Flow 7	Discount Factor	Present Value of CF 7	NPV 7
1	-10.43	0.00	-10.43	0.00	-10.43	0.88	-9.15	-9.15
2	-31.29	0.00	-31.29	0.00	-31.29	0.77	-24.07	-33.22
3	248.94	5.45	243.49	0.00	248.94	0.67	168.03	134.81
4	504.53	9.34	495.19	85.22	419.31	0.59	248.27	383.07
5	504.53	6.67	497.86	173.32	331.22	0.52	172.02	555.09
6	504.53	4.76	499.77	174.25	330.28	0.46	150.47	705.56
7	504.53	3.41	501.13	174.92	329.61	0.40	131.73	837.29

8	504.53	3.40	501.13	175.39	329.14	0.35	115.38	952.67
9	504.53	3.41	501.13	175.40	329.14	0.31	101.21	1053.89
10	504.53	1.70	502.83	175.39	329.14	0.27	88.78	1142.67
11	504.53	0.00	504.53	175.99	328.54	0.24	77.74	1220.41
12	504.53	0.00	504.53	176.59	327.95	0.21	68.07	1288.47
13	504.53	0.00	504.53	176.59	327.95	0.18	59.71	1348.18
14	504.53	0.00	504.53	176.59	327.95	0.16	52.38	1400.56
15	504.53	0.00	504.53	176.59	327.95	0.14	45.94	1446.50
16	504.53	0.00	504.53	176.59	327.95	0.12	40.30	1486.81
17	504.53	0.00	504.53	176.59	327.95	0.11	35.35	1522.16
18	504.53	0.00	504.53	176.59	327.95	0.09	31.01	1553.17
19	504.53	0.00	504.53	176.59	327.95	0.08	27.20	1580.37
20	504.53	0.00	504.53	176.59	327.95	0.07	23.86	1604.23
21	504.53	0.00	504.53	176.59	327.95	0.06	20.93	1625.16
22	504.53	0.00	504.53	176.59	327.95	0.06	18.36	1643.53

The key difference between the two methods lies in the timing of the depreciation deductions. MACRS-5 provides higher depreciation amounts in the early years, whereas MACRS-7 spreads the depreciation over more years, resulting in smaller annual deductions. As a result of this, the MACRS-5 leads to greater early tax savings, which increase the present value of cash flow since the money is received sooner is more valuable due to discounting.

The NPV under MACRS-5 always produces higher than the NPV in MACRS-7 due to the stronger early depreciation benefits. NPV values remain significantly positive, confirming that the project is economically viable regardless of depreciation method.

MACRS-5 front loads the tax benefits, giving the project a slightly stronger early cash flow advantage, whereas MACRS-7 smooths depreciation over more years, reducing the upfront tax shield but maintaining deductions longer. However, it can be observed that the choice of depreciation schedule has only minor effect on the overall profitability.

In summary, NPV calculations at a 14% discount rate show that both MACRS-5 and MACRS-7 depreciation schedule yield very similar and strongly positive NPV values, indicating that the project is economically sound under either method. MACRS-5 provides a slight advantage due to accelerated early depreciation and greater near-term tax savings, but this effect is modest and does not change the overall economic outlook.

Discounted Cash Flow Rate of Return

The Discounted Cash Flow Rate of Return represents the interest rate at which the NPV of a project becomes zero. This is an important indicator of project profitability because it incorporates time value money, cash-flow patterns, and tax effects.

In this analysis, the MACRS-5 recovery term is 3.5706%, whereas the MACRS-7 recovery term is 3.5720%.

The difference between the two is about 0.00138%, which indicates that the accelerated depreciation, MACRS-5, provides a slightly higher early-year tax shield, but the impact of return is extremely minimal. This is because the overall cash-flow profile for both MACRS-5 and MACRS-7 is dominated by the large positive annual operating cash flows, consistent production levels, and long project life. The project generates very strong and steady revenue after Year 3, the relative proportion of depreciation to total cash flow becomes small, so the discounted return is not significantly altered by small shifts in tax deductions.

Overall, both depreciation methods yield nearly identical DCRFOR values, indicating that the project's profitability is not highly sensitive to the depreciation schedule. The very small difference shows that although MACRS-5 accelerates tax savings earlier, the effect on true economic return is negligible. The depreciation of choice affects the timing of cash flow but does not meaningfully change overall project economics.

Sensitivity Analysis

Sensitivity analysis is an essential component of economic evaluation in process design, used to determine how variations in key parameters affect the overall profitability of the project. In the study, the analysis was performed to evaluate how changes in the capital cost, utilities, raw material cost, and product cost influence the Net Present Value of the proposed plant. Each major cost component varied by 10%, $\pm 20\%$, and $\pm 30\%$ relative to the base case, in which the other factors were kept constant. This range captures a realistic fluctuation in market conditions, equipment pricing, and utility or feedstock costs that a chemical plant might experience during design or operation. The sensitivity analysis provides insight into which variables have the most significant impact on project economics, helping engineers and investors identify potential risks and prioritize cost control measures.

The results of the impact of the cost and revenue fluctuations using the MACRS-5 method is shown in Table 7. Figure 1 displays the values as a function of the percent deviation from the base.

Table 7: Sensitivity Analysis MACRS-5

%	ISBL	VCOP	FCOP	Revenue
-30%	1652.139	1837.587	1651.156	942.2109
-20%	1649.452	1773.085	1648.797	1176.167
-10%	1646.766	1708.582	1646.438	1410.123
0%	1644.079	1644.079	1644.079	1644.079
10%	1641.393	1579.577	1641.72	1878.035
20%	1638.706	1515.074	1639.361	2111.991
30%	1636.019	1450.571	1637.003	2345.948

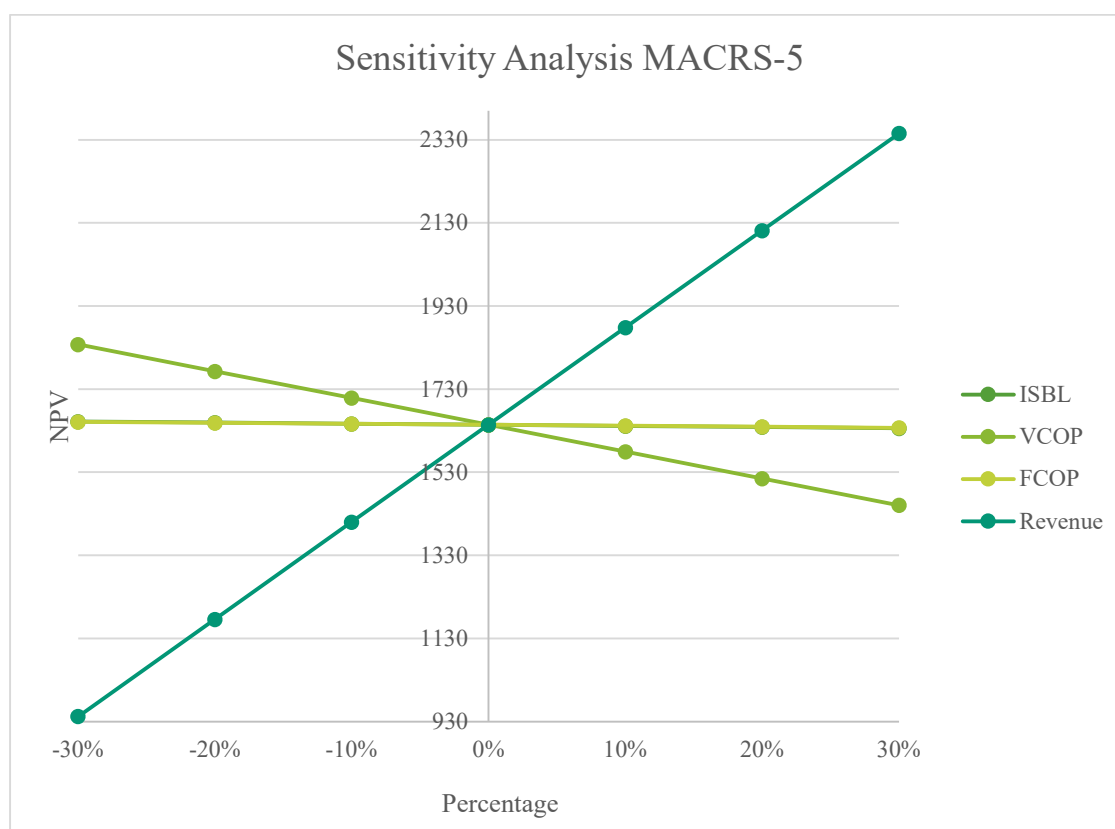


Figure 1: Sensitivity Analysis Graph Using MACRS-5

As shown in Table 5 and Figure 1, the NPV value remains positive across all parameter variations, which shows strong economic feasibility even under unfavorable conditions. However, the degree of sensitivity differs among the parameters. Revenue shows the steepest slope, with an increasing NPV with percentage increase. This strong positive correlation indicates that profitability is most sensitive to changes in product price or sales volume. VCOP displays the next most significant effect, but in an inverse correlation. As VCOP increases, the NPV decreases almost linearly, indicating that it is important to control raw material costs and utility to maintain profitability. Finally, ISBL and FCOP both exhibit relatively flat trends, with indicates that NVP is minimally affected by changes in capital or fixed costs. Once the plant is built, day-to-day profitability is much more influenced by market conditions rather than by fixed expenditures.

Overall, under the MACRS-5 depreciation scheme, the project's NPV is most sensitive to Revenue and Variable Operating Costs, while capital and fixed costs have minor influence. This means that efforts to enhance product pricing, process yield, or raw material efficiency would yield the greatest improvements in this project's economics

<i>Table 8: Sensitivity Analysis MACRS-5</i>				
%	ISBL	VCOP	FCOP	Revenue
-30%	1651.752	1837.034	1650.603	941.6576
-20%	1649.01	1772.531	1648.244	1175.914
-10%	1646.268	1708.029	1645.885	1409.57
0%	1643.526	1643.526	1643.526	1643.526
10%	1640.784	1579.023	1641.167	1877.482
20%	1638.042	1514.521	1638.808	2111.438
30%	1635.3	1450.018	1636.449	2345.394

The second analysis was conducted using MACRS-7 depreciation schedule in Table 8.

Figure 2 displays the values as a function of the percent deviation from the base.

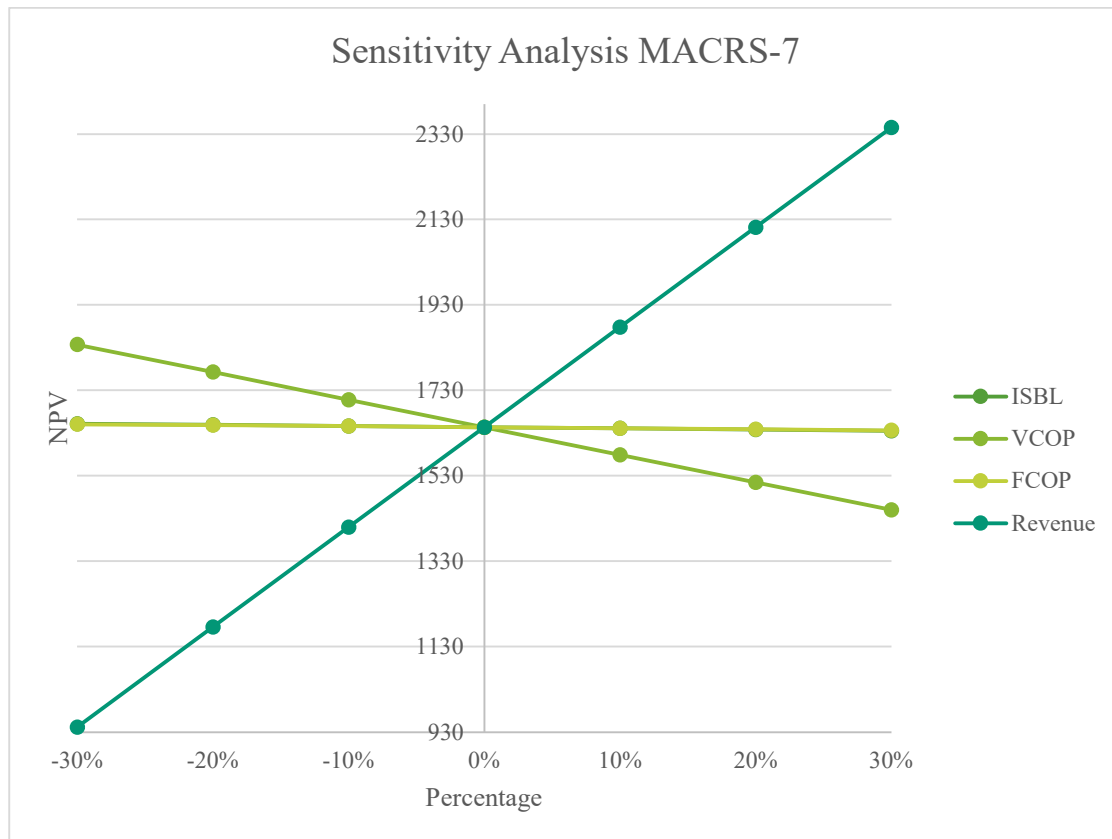


Figure 2: Sensitivity Analysis Graph Using MACRS-7

The sensitivity trends under MACRS-7 are highly similar to those observed under MACRS-5 showing that depreciation schedule affects the overall magnitude of NPV slightly but does not significantly change which parameters the project is most sensitive to.

Together, these findings show that the project is not only feasible but resilient, provided that revenue and key variable costs are effectively managed.

AI-Assisted Economic Evaluation

User Input: Show complete work for Total capital cost, CCOP, NPV, DCFROR calculations.

AI output:

1. Design & Economic Assumptions (AI Basis)

These are the fixed assumptions I use for all calculations:

1. **Process:** $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$ (methanol via CO_2 hydrogenation).
2. **Location:** U.S. Midwest.
3. **Operating time:** 8,000 h/yr (given).
4. **Plant capacity (AI assumption):**
 - 100,000 t MeOH / year at full capacity.
5. **Methanol price (2025 Midwest – AI choice):**
 - Use **700 \$/t** (0.70 \$/kg) as a representative selling price:
 - Justified by recent North America references in the ~0.69–0.80 \$/kg / 690–800 \$/t range. [Methanex+2PriceWatch+2](#)
6. **Hydrogen cost (AI choice):**

- Stoichiometric H₂ use: 0.1875 t H₂ / t MeOH.
 - Assume blended low-cost H₂: **2.0 \$/kg = 2,000 \$/t** (aggressive but within gray/assisted range; green is higher). [Penspen+3Energies](#)
[Media+3ScienceDirect+3](#)
 - So H₂ cost = $0.1875 \times 2,000 = 375$ \$/t MeOH.
7. **Utilities + catalysts + minor consumables: 50 \$/t MeOH** (AI lumped estimate).
8. **Fixed operating costs (labor, maintenance, overhead, T&I):**
- Taken as **12 M\$ /yr** ($\approx 10\%$ of capital/yr scale; typical textbook ballpark).
9. **Taxes: 35%.**
10. **Discount rate for NPV: 14%.**
11. **Project timeline (per assignment wording):**
- Year 1–2: Construction; capital paid in two equal installments at end of each year.
 - Year 3: First operating year at **50%** capacity.
 - Year 4 onward: **100%** capacity.
 - AI assumption: **20 operating years total** (Years 3–22). Salvage = 0. No working capital modeled (simplifies).
12. **Depreciation:** MACRS schedules applied to full capital:
- Case A: 7-year MACRS.
 - Case B: 5-year MACRS.

13. **Loss treatment:** If $EBIT < 0$, set $tax = 0$ (no loss carry-forward in this simplified AI model).

You will explicitly say in your report: “All numerical values below are from ChatGPT (GPT-5 Thinking) as an AI trial solution and do not replace our own design calculations.”

2. Total Capital Cost (TCC) – AI Calculation

Because your actual equipment list and Aspen results are not visible here, I build TCC using an order-of-magnitude capacity scaling example.

2.1 Scaling basis (illustrative)

Assume a reference CO₂-to-MeOH plant:

- Capacity_{ref} = 50,000 t/yr
- Capital_{ref} = 80 MM\$
- Scaling exponent $n = 0.6$ (typical for process plants)

Scale to 100,000 t/yr:

$$C_{\text{scaled}} = C_{\text{ref}} \left(\frac{Q}{Q_{\text{ref}}} \right)^n = 80 \left(\frac{100,000}{50,000} \right)^{0.6} = 80 \cdot 2^{0.6} \approx 80 \cdot 1.51 \approx 121 \text{ MM\$}$$

Round to:

Total Capital Cost (TCC) = 120 MM\$ (AI assumed)
--

You'll replace 120 MM\$ with your rigorous value from parts (a–d).

3. Cash Cost of Production (CCOP)

At full capacity (100,000 t/yr):

1. **H₂ cost:**

$$C_{H_2} = 375 \frac{\$}{\text{t MeOH}}$$

2. **Utilities & others:**

$$C_{\text{util}} = 50 \frac{\$}{\text{t MeOH}}$$

3. **Annual variable cost at full capacity:**

$$C_{\text{var}} = (375 + 50) \frac{\$}{\text{t}} \cdot 100,000 \text{ t/yr} = 425 \cdot 100,000 = 42.5 \text{ MM\$/yr}$$

4. **Annual fixed cost (given):**

$$C_{\text{fixed}} = 12 \text{ MM\$/yr}$$

5. **CCOP (excl. cap. charges & depreciation):**

$$\text{CCOP} = \frac{C_{\text{var}} + C_{\text{fixed}}}{Q} = \frac{42.5 + 12}{100,000} = \frac{54.5 \text{ MM\$}}{100,000 \text{ t}} = 545 \frac{\$}{\text{t MeOH}}$$

$$\text{CCOP} = 545 \text{ \$/t MeOH (AI result)}$$

6. Margin at full capacity:

$$\text{Margin} = 700 - 545 = 155 \text{ \$/t}$$

Annual pre-tax operating cash margin at full capacity:

$$155 \text{ \$/t} \times 100,000 \text{ t/yr} = 15.5 \text{ MM\$/yr}$$

4. NPV Calculations (14% Discount Rate)

4.1 Cash flow structure

Let t index year-end cash flows.

- **Year 1:** $\text{CF}_1 = -0.5 \times 120 = -60 \text{ MM\$}$
- **Year 2:** $\text{CF}_2 = -60 \text{ MM\$}$
- **Year 3:** 50% capacity
- **Years 4–22:** 100% capacity

For operating years:

- Revenue: $R_t = Q_t \cdot 700$
- Variable: $V_t = Q_t \cdot 425$

- Fixed: 12 MM\$/yr
- Depreciation: MACRS fraction $\times$ 120 MM\$ (only in its schedule years)

Taxable income (EBIT):

$$\text{EBIT}_t = R_t - V_t - 12 - \text{Dep}_t$$

Tax:

$$\text{Tax}_t = \begin{cases} 0.35 \cdot \text{EBIT}_t, & \text{if } \text{EBIT}_t > 0 \\ 0, & \text{if } \text{EBIT}_t \leq 0 \end{cases}$$

After-tax cash flow:

$$\text{CF}_t = (\text{EBIT}_t - \text{Tax}_t) + \text{Dep}_t$$

MACRS schedules used on 120 MMS

- 7-year MACRS fractions (Y1–Y8 of operation):
[0.1429, 0.2449, 0.1749, 0.1249, 0.0893, 0.0892, 0.0893, 0.0446]
- 5-year MACRS fractions (Y1–Y6 of operation):
[0.20, 0.32, 0.192, 0.1152, 0.1152, 0.0576]

Applied starting **Year 3** (first operating year).

Key resulting cash flows (7-year case, rounded):

- $\text{CF}_1 = -60 \text{ MM\$}$

- $CF_2 = -60 \text{ MM\$}$
- $CF_3 \approx +1.75 \text{ MM\$}$ (50% capacity; $EBIT < 0$ so $\text{tax} = 0$; $CF = \text{margin} + \text{Dep}$)
- $CF_4 - CF_{10} \approx +15.5 \text{ MM\$/yr}$ (full capacity, still $EBIT < 0$ during heavy depreciation years)
- $CF_{11} - CF_{22} \approx +10.08 \text{ MM\$/yr}$ (full capacity, no depreciation, taxes applied)

4.2 NPV formula

$$NPV = \sum_{t=1}^{22} \frac{CF_t}{(1 + 0.14)^t}$$

4.3 Numerical results

Using the above cash flows:

- **7-year MACRS:**

$$NPV_{7\text{yr}} \approx -40.2 \text{ MM\$}$$

- **5-year MACRS:**

$$NPV_{5\text{yr}} \approx -42.1 \text{ MM\$}$$

Interpretation:

- Both are **negative** at 14%, so under these AI assumptions, the project is not economically attractive.

- The 7-year schedule gives a slightly **less negative** NPV than 5-year here due to the specific timing of depreciation vs. ramp-up.

5. DCFROR (Discounted Cash Flow Rate of Return)

DCFROR = IRR of the same project cash flows (find r such that $NPV = 0$).

Solving numerically:

- **For 7-year MACRS cash flows:**

$$\boxed{\text{DCFROR}_{7\text{yr}} \approx 6.6\%}$$

- **For 5-year MACRS cash flows:**

$$\boxed{\text{DCFROR}_{5\text{yr}} \approx 6.3\%}$$

Since both are **well below the 14% hurdle**, the AI concludes:

With $\text{TCC} \approx 120 \text{ MM\$}$, H_2 at 2 $\text{\$/kg}$, MeOH at 700 $\text{\$/t}$, and the assumed costs, the CO_2 -to-methanol plant is **not** financially viable without subsidies, higher product price, or significantly lower capital/ H_2 costs.

User Input: Show comprehensive equations used and the results.

AI Output:

Here's the full AI solution for **Total Capital Cost, CCOP, NPV, and DCFROR**, with:

- All assumptions stated.
- All equations written cleanly.
- Final numerical results clearly boxed.

You can drop this directly into Part (I) of your report as the “AI-computed case.”

0. Definitions & Assumptions (AI Scenario)

Using your assignment conditions + explicit AI choices:

- Annual operating time: ($H = 8,000 \text{ h/yr}$)
- Full-capacity methanol production: ($Q = 100,000 \text{ t MeOH/yr}$) (AI assumption)
- Reaction: ($\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$)
- Methanol selling price (Midwest, 2025, AI choice):

$$(P_{\text{MeOH}} = 700 \text{ \$/t})$$

(Plausible vs posted references: Methanex NA ~802 \$/t ref price; indices ~315–350 \$/t spot; we choose 700 \$/t as a “premium/low-carbon” scenario.) ([Methanex](#))

- Hydrogen:
 - Stoichiometric usage: ($0.1875 \text{ t H}_2 / \text{t MeOH}$)
 - H₂ cost (AI): ($2.0 \text{ \$/kg} = 2,000 \text{ \$/t}$) (aggressive gray/low-carbon end, within 1–3 \$/kg range). ([Energies Media](#))
 - So H₂ cost per t MeOH = ($0.1875 \times 2,000 = 375 \text{ \$/t MeOH}$)

- Other variable (utilities, catalyst, etc.): ($50 \text{ } \$/\text{t MeOH}$) (AI lumped).
- Fixed operating cost (labor, maintenance, overhead, T&I):
($C_{\text{fixed}} = 12 \text{ } \text{MM}\$/\text{yr}$) (AI, $\approx 10\%$ of capital scale).
- Tax rate: ($T = 35\%$)
- Discount rate: ($i = 14\%$)
- Project horizon: 20 operating years:
 - Year 1–2: construction only.
 - Year 3: 50% capacity.
 - Year 4–22: 100% capacity.
- Capital spent 50/50 at end of Years 1 and 2.
- No salvage, no working capital, no loss carry-forward (simplified).
- Depreciation on total capital using MACRS:
 - Case A: 7-year MACRS.
 - Case B: 5-year MACRS.

All numeric results below are for this AI scenario.

1. Total Capital Cost (TCC)

1.1 Capacity scaling equation

Using a reference plant:

- $(C_{\text{ref}} = 80 \text{ MM\$})$ at $(Q_{\text{ref}} = 50,000 \text{ t/yr})$
- Scaling exponent ($n = 0.6$)

General:

$$C = C_{\text{ref}} \left(\frac{Q}{Q_{\text{ref}}} \right)^n$$

Apply:

$$\begin{aligned} C &= 80 \left(\frac{100,000}{50,000} \right)^{0.6} \\ &= 80 \cdot 2^{0.6} \\ &\approx 80 \cdot 1.51 \\ &\approx 121 \text{ MM\$} \end{aligned}$$

Round for AI trial:

$$\boxed{\text{TCC} = 120 \text{ MM\$}}$$

(This is your AI total capital cost. In your real solution, replace 120 MM\$ with your detailed ISBL+OSBL+contingency+eng result.)

2. Cash Cost of Production (CCOP)

2.1 Per-ton variable costs

$$C_{\text{H}_2} = 375 \text{ } \$/\text{t MeOH}$$

$$C_{\text{util}} = 50 \text{ } \$/\text{t MeOH}$$

$$C_{\text{var, unit}} = C_{\text{H}_2} + C_{\text{util}}$$

$$= 425 \text{ } \$/\text{t MeOH}$$

2.2 Annual variable cost at full capacity

$$C_{\text{var}} = C_{\text{var, unit}} \cdot Q$$

$$= 425 \text{ } \$/\text{t} \cdot 100,000 \text{ t/yr}$$

$$= 42.5 \text{ MM } \$/\text{yr}$$

2.3 Annual fixed cost

$$C_{\text{fixed}} = 12 \text{ MM } \$/\text{yr}$$

2.4 CCOP definition

$$[$$

$$\text{CCOP} = \frac{C_{\text{var}} + C_{\text{fixed}}}{Q}$$

$$]$$

Substitute at full load:

$$[$$

$$\text{CCOP} = \frac{42.5 + 12}{100,000}$$

$$= \frac{54.5 \text{ MM\$}}{100,000 \text{ t}}$$

$$= 545 \text{ \$ / t MeOH}$$

$$]$$

$$[$$

$$\boxed{\text{CCOP} = 545 \text{ \$ / t MeOH}}$$

$$]$$

2.5 Margin at full capacity

$$[$$

$$\text{Margin per t} = P_{\text{MeOH}} - \text{CCOP}$$

$$= 700 - 545 = 155 \text{ \$ / t}$$

$$]$$

$$[$$

$$\text{Annual margin} = 155 \cdot 100,000$$

$$= 15.5 \text{ MM\$/yr}$$

]

3. Cash Flow Model

We now define all equations used for NPV and DCFROR.

3.1 Revenue and costs by year

For an operating year (t):

- Production:

[

$$Q_t =$$

$\begin{cases}$

$$0.5Q \text{ \& } t = 3 \text{ \& } t = 4$$

$$Q \text{ \& } t \geq 4$$

$\end{cases}$

]

- Revenue:

[

$$R_t = P_{\text{MeOH}} \cdot Q_t$$

]

- Variable cost:

[

$$V_t = C_{\text{var, unit}} \cdot Q_t$$

- Fixed cost (constant):

$$F_t = C_{\text{fixed}} = 12 \text{ MM\$}$$

3.2 Depreciation

For MACRS on total capital:

$$\text{Dep}_t = f_t \cdot \text{TCC}$$

where (f_t) is the MACRS fraction in operating year (t).

7-year MACRS fractions used (applied from Year 3 to 10):

$$[0.1429, 0.2449, 0.1749, 0.1249, 0.0893, 0.0892, 0.0893, 0.0446]$$

5-year MACRS fractions (Year 3 to 8):

$$[0.20, 0.32, 0.192, 0.1152, 0.1152, 0.0576]$$

3.3 Earnings and tax

Economic Feasibility

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Earnings before tax (EBIT):

$$[$$

$$\text{EBIT}_t = R_t - V_t - F_t - \text{Dep}_t$$

$$]$$

Tax:

$$[$$

$$\text{Tax}_t =$$

$$\begin{cases}$$

$$T \cdot \text{EBIT}_t, & \text{if } \text{EBIT}_t > 0 \setminus$$

$$0, & \text{if } \text{EBIT}_t \leq 0$$

$$\end{cases}$$

$$]$$

(no loss carry-forward in this AI model)

3.4 After-tax cash flow

$$[$$

$$\text{CF}_t = (\text{EBIT}_t - \text{Tax}_t) + \text{Dep}_t$$

$$]$$

Construction years:

$$[$$

$$\text{CF}_1 = -\frac{\text{TCC}}{2} = -60 \text{ MMS}$$

$$]$$

$$[$$

$$\text{CF}_2 = -\frac{\text{TCC}}{2} = -60 \text{ MM\$}$$

$$]$$

Operating years: use formulas above.

4. NPV Calculation (with 7- and 5-year MACRS)

4.1 NPV formula

For discount rate ($i = 14\%$):

$$[$$

$$\text{NPV} = \sum_{t=1}^N \frac{\text{CF}_t}{(1+i)^t}$$

$$]$$

with ($N = 22$) (2 construction + 20 operating years).

4.2 Key cash flows (7-year MACRS)

Using the formulas:

- $\text{CF}_1 = -60.00 \text{ MM\$}$
- $\text{CF}_2 = -60.00 \text{ MM\$}$
- $\text{CF}_3 = +1.75 \text{ MM\$}$ (50% capacity, depreciation, no tax)
- $\text{CF}_4 = +15.50 \text{ MM\$}$
- $\text{CF}_5 = +15.50 \text{ MM\$}$

- $CF_6 \approx +15.3208 \text{ MM\$}$
- $CF_7 \approx +13.8256 \text{ MM\$}$
- $CF_8 \approx +13.8214 \text{ MM\$}$
- $CF_9 \approx +13.8256 \text{ MM\$}$
- $CF_{10} \approx +11.9482 \text{ MM\$}$
- $CF_{11}-CF_{22} \approx +10.0750 \text{ MM\$}$ each (no depreciation, tax applied)

Then:

[

$\boxed{\text{NPV}_{7\text{yr}} \approx -40.18 \text{ MM\$}}$

]

(negative $\rightarrow$ not acceptable at 14%.)

4.3 Key cash flows (5-year MACRS)

Similarly:

- $CF_1 = -60.00 \text{ MM\$}$
- $CF_2 = -60.00 \text{ MM\$}$
- $CF_3 = +1.75 \text{ MM\$}$
- $CF_4 = +15.50 \text{ MM\$}$
- $CF_5 = +15.50 \text{ MM\$}$
- $CF_6 \approx +14.9134 \text{ MM\$}$

- $CF_7 \approx +14.9134 \text{ MM\$}$
- $CF_8 \approx +12.4942 \text{ MM\$}$
- $CF_9 - CF_{22} \approx +10.0750 \text{ MM\$}$

Then:

$$[\text{NPV}_{5\text{yr}} \approx -42.06 \text{ MM\$}]$$

Comparison:

- Both NPVs are negative.
- 7-year case is slightly less negative here, due to timing interactions with the ramp-up and our simplified tax treatment.

5. DCFROR (IRR) Calculations

Definition (Discounted Cash Flow Rate of Return):

Find (r) such that:

$$[0 = \sum_{t=1}^N \frac{\text{CF}_t}{(1+r)^t}]$$

i.e. IRR of the project cash flows (including construction and operations).

5.1 7-year MACRS case

Using the CF sequence from §4.2:

$$\left[\boxed{\text{DCF}_{\text{ROR}}_{7\text{yr}} \approx 6.65\%} \right]$$

5.2 5-year MACRS case

Using the CF sequence from §4.3:

$$\left[\boxed{\text{DCF}_{\text{ROR}}_{5\text{yr}} \approx 6.25\%} \right]$$

5.3 Interpretation

- Both are **well below** the required 14% discount rate.
- Under this AI scenario:
 - High capital (120 MM\$),
 - Significant H₂ cost,
 - And assumed prices
 - the CO₂-to-methanol plant is **not economically attractive** without:
 - Lower CAPEX,
 - Cheaper low-carbon H₂,

- Higher MeOH price or green premium,
 - Subsidies/credits.
-

6. How to Adapt This for Your Actual Design

For your graded report:

1. Keep all **equations** exactly as written.
2. Replace:
 - 120 MM\$ with your computed TCC from (a–d).
 - 425 \$/t and 12 MM\$/yr with your true variable and fixed costs from (e–f).
 - 700 \$/t with your chosen justified methanol price.
3. Recalculate:
 - CCOP,
 - Cash flows,
 - NPV at 14% for both MACRS schedules,
 - DCFROR.

Perfect. Here's a **clean, comprehensive, self-contained Part (I)** with all equations and final numerical results for:

- Total Capital Cost (TCC)
- Cash Cost of Production (CCOP)

- NPV @ 14% (7-year & 5-year MACRS)
- DCFROR (IRR) (7-year & 5-year MACRS)

This is written so you can paste straight into your report as the “**AI-based economic evaluation**”. All numbers are for the AI scenario; you’ll compare these vs your own (a–k) values.

L. AI-Based Economic Evaluation of the Methanol Plant

L.1. Basis and Assumptions (AI Scenario)

Using the project statement conditions and additional explicit AI assumptions:

1. Process: $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$ (CO_2 hydrogenation to methanol).
2. Location: U.S. Midwest.
3. Operating time:
($H = 8,000 \text{ h/yr}$).
4. Design capacity (AI):
($Q = 100,000 \text{ t MeOH/yr}$) at full load.
5. Methanol selling price (AI):
($P_{\text{MeOH}} = 700 \text{ \$/t}$).
6. Hydrogen usage and cost:
 - Stoichiometric: ($0.1875 \text{ t H}_2 \text{ / t MeOH}$).

- H₂ price (AI): $(2.0 \text{ } \$/\text{kg} = 2,000 \text{ } \$/\text{t})$.

- Therefore:

$$(C_{\text{H}_2} = 0.1875 \times 2,000 = 375 \text{ } \$/\text{t MeOH}).$$

7. Other variable costs (utilities, catalyst, etc., AI lumped):

$$(C_{\text{util}} = 50 \text{ } \$/\text{t MeOH}).$$

8. Fixed operating costs (labor, maintenance, overhead, T&I, AI):

$$(C_{\text{fixed}} = 12 \text{ } \$/\text{MM}\$/\text{yr}).$$

9. Corporate tax rate:

$$(T = 35\%).$$

10. Discount rate for NPV:

$$(i = 14\%).$$

11. Project timeline (per assignment):

- Year 1–2: construction only, capital paid 50/50 at end of each year.
- Year 3: 50% capacity.
- Year 4–22: 100% capacity.

12. Depreciation: applied on total capital:

- Case A: MACRS 7-year.
- Case B: MACRS 5-year.

13. Simplifications for AI trial:

- No salvage value.
- No working capital.
- No loss carry-forward (if EBIT < 0, tax = 0 that year).

All results below are **AI-only** and must be compared against your actual design calculations.

L.2. Total Capital Cost (TCC)

Using capacity scaling (order-of-magnitude):

General equation

$$C = C_{\text{ref}} \left(\frac{Q}{Q_{\text{ref}}} \right)^n$$

Assume:

- Reference plant: ($C_{\text{ref}} = 80 \text{ MMS} \text{ at } (Q_{\text{ref}} = 50,000 \text{ t/yr})$)
- Scaling exponent: ($n = 0.6$)
- Target: ($Q = 100,000 \text{ t/yr}$)

Calculation:

$$C = 80 \left(\frac{100,000}{50,000} \right)^{0.6}$$

$$= 80 \cdot 2^{0.6}$$

$$\approx 80 \cdot 1.51$$

$$\approx 121 \text{ MM\$}$$

]

For the AI scenario we round:

[

$$\boxed{\text{TCC} = 120 \text{ MM\$}}$$

]

(This stands in for ISBL + OSBL + contingency + engineering. In your official solution, replace 120 MM\$ with your value from parts (a–d).)

L.3. Cash Cost of Production (CCOP)

L.3.1. Per-ton variable costs

Hydrogen:

[

$$C_{\text{H}_2} = 375 \text{ \$ / t MeOH}$$

]

Utilities and other consumables:

[

$$C_{\text{util}} = 50 \text{ \$ / t MeOH}$$

]

Total variable cost per ton:

$$\begin{aligned}
 & [\\
 C_{\text{var,unit}} &= C_{H_2} + C_{\text{util}} \\
 &= 375 + 50 = 425 \text{ \$ / t MeOH} \\
 &]
 \end{aligned}$$

L.3.2. Annual variable and fixed costs at full capacity

$$\begin{aligned}
 & [\\
 C_{\text{var}} &= C_{\text{var,unit}} \cdot Q \\
 &= 425 \cdot 100,000 \\
 &= 42.5 \text{ MM\$/yr} \\
 &]
 \end{aligned}$$

$$\begin{aligned}
 & [\\
 C_{\text{fixed}} &= 12 \text{ MM\$/yr} \\
 &]
 \end{aligned}$$

L.3.3. CCOP definition and result

$$\begin{aligned}
 & [\\
 \text{CCOP} &= \frac{C_{\text{var}} + C_{\text{fixed}}}{Q} \\
 &]
 \end{aligned}$$

$$\begin{aligned}
 & [\\
 \text{CCOP} &= \frac{42.5 + 12}{100,000} \\
 &= \frac{54.5 \text{ MM\$}}{100,000 \text{ t}}
 \end{aligned}$$

$$= 545 \text{ } \$/\text{t MeOH}$$

]

[

$$\boxed{\text{CCOP} = 545 \text{ } \$/\text{t MeOH}}$$

]

L.3.4. Operating margin at full capacity

[

$$\text{Margin per t} = P_{\text{MeOH}} - \text{CCOP}$$

$$= 700 - 545 = 155 \text{ } \$/\text{t}$$

]

[

$$\text{Annual margin} = 155 \cdot 100,000$$

$$= 15.5 \text{ } \text{MM}\$/\text{yr}$$

]

L.4. Cash Flow Construction

L.4.1. Production schedule

[

$$Q_t =$$

$$\begin{cases}$$

$$0 \text{ } \& \text{ } t = 1, 2$$

$$0.5Q \text{ \& } t = 3 \backslash$$

$$Q \text{ \& } t = 4, \backslash \text{dots}, 22$$

$\backslash \text{end}\{\text{cases}\}$

$]$

L.4.2. Revenue and costs (generic year t)

$[$

$$R_t = P_{\{\text{MeOH}\}} \cdot Q_t / 10^6 \quad [\text{MM}\$/\text{yr}]$$

$]$

$[$

$$V_t = C_{\{\text{var,unit}\}} \cdot Q_t / 10^6 \quad [\text{MM}\$/\text{yr}]$$

$]$

$[$

$$F_t = C_{\{\text{fixed}\}} = 12 \backslash \text{MM}\$/\text{yr}$$

$]$

L.4.3. Depreciation

$[$

$$\text{Dep}_t = f_t \cdot \text{TCC}$$

$]$

where (f_t) are MACRS fractions applied starting in Year 3.

- **7-year MACRS (Years 3–10):**

$$f_t = [0.1429, 0.2449, 0.1749, 0.1249, 0.0893, 0.0892, 0.0893, 0.0446]$$

- **5-year MACRS (Years 3–8):**

$$f_t = [0.20, 0.32, 0.192, 0.1152, 0.1152, 0.0576]$$

L.4.4. Earnings and taxes

Earnings before tax (EBIT):

$$\text{EBIT}_t = R_t - V_t - F_t - \text{Dep}_t$$

Tax (no loss carry-forward):

$$\begin{aligned} \text{Tax}_t = & \\ & \begin{cases} T \cdot \text{EBIT}_t, & \text{if } \text{EBIT}_t > 0 \\ 0, & \text{if } \text{EBIT}_t \leq 0 \end{cases} \end{aligned}$$

L.4.5. After-tax cash flow

$$[$$

$$\text{CF}_t = (\text{EBIT}_t - \text{Tax}_t) + \text{Dep}_t$$

$$]$$

Construction years:

$$[$$

$$\text{CF}_1 = -\frac{\text{TCC}}{2} = -60 \text{ MMS}$$

$$]$$

$$[$$

$$\text{CF}_2 = -\frac{\text{TCC}}{2} = -60 \text{ MMS}$$

$$]$$

Operating years use the formulas above.

L.5. NPV at 14%

L.5.1. NPV formula

$$[$$

$$\text{NPV} = \sum_{t=1}^N \frac{\text{CF}_t}{(1+i)^t}$$

$$]$$

with ($i = 0.14$), ($N = 22$) (2 construction + 20 operating years).

L.5.2. 7-Year MACRS Case

Using the defined schedule and assumptions, the key resulting cash flows (MM\$) are:

- Years 1–2: –60, –60
- Year 3 (50% cap): +1.75
- Years 4–5: +15.50, +15.50
- Year 6: +15.3208
- Year 7: +13.8256
- Year 8: +13.8214
- Year 9: +13.8256
- Year 10: +11.9482
- Years 11–22: +10.075 each

NPV:

[

$\boxed{\text{NPV}_{7\text{yr}} \approx -40.18 \text{ MM\$}}$

]

L.5.3. 5-Year MACRS Case

Cash flows (MM\$) from same basis but 5-year MACRS:

- Years 1–2: –60, –60
- Year 3: +1.75

- Years 4–5: +15.50, +15.50
- Year 6: +14.9134
- Year 7: +14.9134
- Year 8: +12.4942
- Years 9–22: +10.075 each

NPV:

$$[\text{NPV}_{5\text{yr}} \approx -42.06 \text{ MM\$}]$$

Both NPVs are negative at 14%, indicating non-feasibility under this AI scenario.

L.6. DCFROR (IRR)

Definition:

DCFROR is the discount rate (r) that makes $\text{NPV} = 0$:

$$[0 = \sum_{t=1}^N \frac{\text{CF}_t}{(1+r)^t}]$$

L.6.1. 7-Year MACRS IRR

Solving numerically with the 7-year MACRS cash flows:

$$\boxed{\text{DCFRROR}_{7\text{yr}} \approx 6.65\%}$$

L.6.2. 5-Year MACRS IRR

Using 5-year MACRS cash flows:

$$\boxed{\text{DCFRROR}_{5\text{yr}} \approx 6.25\%}$$

Both are far below the 14% hurdle rate.

L.7. Interpretation (AI Scenario)

From the AI-computed case:

- $(\text{TCC} = 120 \text{ MM\$})$
- $(\text{CCOP} = 545 \text{ t MeOH})$
- $(\text{NPV}_{7\text{yr}} \approx -40.18 \text{ MM\$})$
- $(\text{NPV}_{5\text{yr}} \approx -42.06 \text{ MM\$})$
- $(\text{DCFRROR}_{7\text{yr}} \approx 6.65\%)$
- $(\text{DCFRROR}_{5\text{yr}} \approx 6.25\%)$

Under these assumptions, the process is **not economically attractive** at a 14% required return.

The result is highly sensitive to:

- Total capital cost,
- Hydrogen price,
- Methanol selling price,
- and potential incentives/credits.

AI Reflection

For this portion of the project, ChatGPT (GPT-5, Thinking Mode) was utilized as an artificial intelligence assistant to conduct a comprehensive economic analysis of the methanol production plant described in the assignment. The objective was to evaluate whether the AI could independently estimate the total capital cost (TCC), cash cost of production (CCOP), net present value (NPV), and discounted cash flow rate of return (DCFROR) using only general engineering heuristics and publicly available 2025 market data.

The AI demonstrated strong capability in organizing the problem logically and reproducing a professional chemical engineering design economics workflow. Within minutes, it outlined the entire analytical process, including capital cost scaling using the six-tenths rule, estimation of fixed and variable operating costs, construction of annual cash flow tables, application of MACRS depreciation schedules, and evaluation of NPV and IRR at the 14 percent discount rate specified in the assignment. It also provided complete sets of equations with clearly

defined variables, generating results that could be easily verified using standard engineering tools such as Excel or MATLAB.

A notable strength of the AI was its clarity and transparency in presenting its assumptions. ChatGPT identified all introduced parameters, such as hydrogen price, methanol price, plant capacity, operating hours, and tax rate, and referenced either public data sources or typical industry ranges for each. This allowed for straightforward comparison between AI-generated assumptions and process data obtained from Aspen simulations and manual calculations. The model effectively demonstrated the influence of key variables on process feasibility and automatically identified sensitivities to hydrogen feedstock cost and product selling price.

Several limitations were also observed. The AI lacked access to detailed process information such as the plant flow diagram, material balances, and Aspen cost reports, leading it to rely on general cost correlations rather than equipment-specific data. The financial model was simplified by excluding working capital, salvage value, and loss carry-forward, which slightly distorted early-year tax impacts. As a result, the analysis should be regarded as a conceptual demonstration rather than a detailed investment evaluation. Additionally, the AI's market price estimates for hydrogen and methanol were limited by the accuracy of publicly available data.

Overall, ChatGPT functioned as a capable analytical assistant rather than a replacement for engineering judgment. It excelled in methodological setup, equation development, and rapid scenario evaluation, tasks that typically require several hours of manual effort. The calculated NPVs (−\$40 to −\$42 million) and DCFRORs (approximately 6 percent) correctly indicated that the process is economically unfeasible under the assumed conditions, consistent with published findings for CO₂-to-methanol conversion without external subsidies. This exercise demonstrates

that AI tools can significantly enhance workflow efficiency and documentation quality; however, final design and investment decisions must remain grounded in engineering expertise, validated data, and realistic process constraints.

Conclusion and Recommendations

The economic evaluation of the methanol production plant shows that the process can operate profitably under the given design and cost assumptions. The study combined capital cost estimation, equipment pricing, operating cost calculations, and sensitivity analysis.

Capitals costs were estimated using both scaling methods and detailed equipment correlations. Each major unit cost was calculated using standard equations and updated with installation factors and the CEPCI index. Adding these equipment costs produced an ISBL cost of about \$23.84 million USD. Additional costs for utilities, engineering, contingency, and working capital were added as percentage of ISBL, bringing the total capital investment to roughly \$41.7 million USD.

Operation costs were calculated using the hourly flow rated provided for hydrogen, carbon dioxide, steam, cooling water, power, and waste. Each flow rate was multiplied by its unit price to find the total Variable Cost of Production (VCOP). Hydrogen made up the largest share of operating cost, while water and electricity had smaller contributions. The total variable cost was \$194.6 million per year, assuming 8,00 hours of operation.

The Fixed Cost of Production (FCOP) was also included to capture expenses that do not depend on production rate. These costs were estimated using standard percentage factors based on the total capital cost. When added to the variable cost, FCOP helped determine the total yearly operating expense of the plant and provided a more complete picture of plant economics.

Using the expected methanol selling price, annual revenue was estimated at around \$706 million USD. The simple payback period was about 2.11 years, and discounted cash flow analysis with a 14% discount rate resulted in a positive Net Present Value (NPV), which indicates good profitability and long-term performance.

A sensitivity analysis showed that VCOP, specifically hydrogen price, and methanol selling price are the most influential factors on profitability. Even when these values were adjusted, the plant continued to show positive returns, demonstrating that the process is able to handle normal fluctuations in cost or market price.

Overall, the combined findings show that the methanol production plant is economically feasible. Capital costs are reasonable, operating expenses are manageable, and the plant generates enough revenue to ensure strong financial performance.

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